FORS 250 / GPHY 285 (Autumn 2025)

Please refer to the syllabus on Canvas for the most up-to-date version of this document. Some minor changes may be implemented over the course of the semester.

Course Syllabus

Instructor

Jonathan Lamont Batchelor *(they/them)*.

Jonathan.Batchelor@UMontana.edu, However, Please use Canvas for all communication about the course.

- Office hours: Tuesday 2pm to 4pm **CHCB 449**
  - By appointment as well

Teaching Assistants

- **Segura, Haley** (haley.segura@umconnect.umt.edu)
  - Office Hours TBD
- **White, Dylan** (dylan1.white@umconnect.umt.edu)
  - Office Hours TBD
- **Benson, Patrick** (patrick.benson@umconnect.umt.edu)
  - Office Hours TBD

Lecture / Lab

Lecture (All Sections)

- **FORS 250 – 00 71142 & GPHY 284 – 00 70936**
  - Tuesday 11:00 - 12:20, Stone 304
  - Thursday 11:00 - 12:20, Stone 304

Lab by section

- **FORS 250 -01 71143**
  - Friday 11:00 - 12:50, Stone 106
- **GPHY 284-02 70937**
  - Friday 12:00 - 13:50, Stone 107
- **FORS 250-03 71144**
  - Friday 13:00 - 14:50, Stone 106
- **GPHY 284-04 70938**
  - Friday 14:00 - 15:50, Stone 107
- **FORS 250-05 71145**
  - Friday 15:00 - 16:50, Stone 106

Description

This course is designed as a practical introduction to the use of Geographic Information Systems (GIS) for storing, retrieving, analyzing and displaying spatial data. It will also cover the history of cartography as an art form and the cartographic conventions of the modern map-making process. Finally, we will review many examples of applied GIS in the private, public, and non-profit sectors.
The course will discuss all the major concepts and theories behind the use of GIS and put these skills into practice with laboratory exercises using GIS software. We will closely examine the proper use of data, both from its appropriate choice given a desired outcome, and to inform a deeper understanding of data manipulation within the “Black Box.” Secondly, the course will address the concepts of proper map design so that our maps more effectively convey the information we intend to impart.

Objectives:

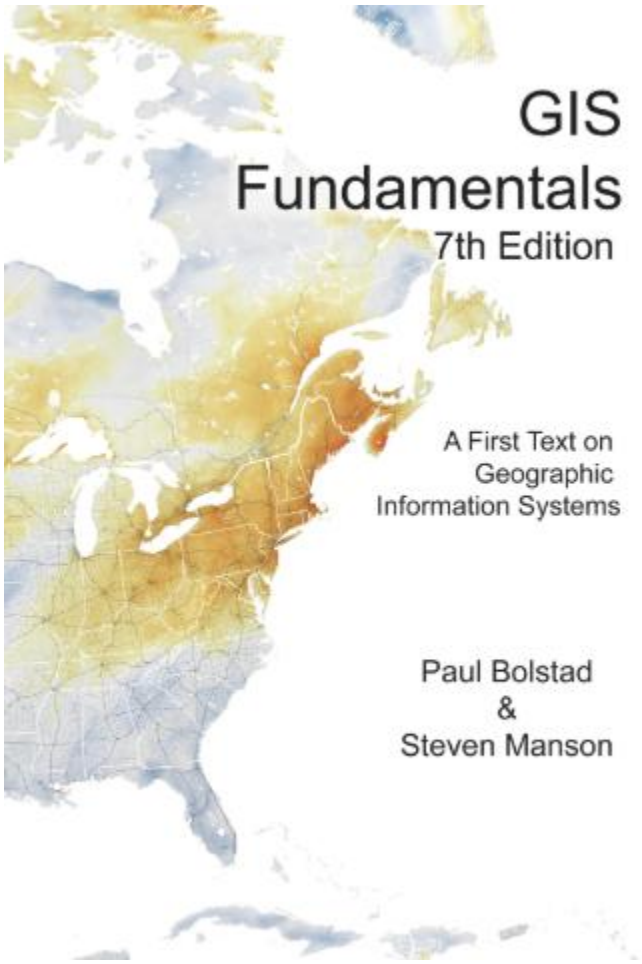
- To learn the foundations of GIS in order to properly and truthfully display spatial data in a cartographic context.
- To understand the structure of spatial data including file associations, attribute tables, metadata, coordinate systems, and projections.
- To become proficient in the construction of geodatabases (GDB) as the fundamental data storage format within all GIS Science.
- To learn basic GIS analysis for answering rudimentary spatial questions using the scientific method.
- To develop software skills in programs used for map production in the modern cartographic workflow.

Learning Outcomes:

1. Understand the core concepts and functions of geographic information systems.

2. Be able to discriminate among different characterizations of the Earth's surface and choose projections that are suitable for specific forestry, geography, and natural resources applications.
3. Recognize and acquire different forms of spatial data, normalize its spatial reference systems, and organize it into a geodatabase (GDB) in preparation for GIS analysis.
4. Be familiar with techniques of spatial analysis applied in forestry, geography, and natural resource management.
5. Be able to produce maps that appropriately display the outcomes of spatial analysis and communicate the results using proper cartographic conventions.

Required Materials:



Bolstad, Paul & Manson, Steven (2022). GIS Fundamentals, 7th Edition
ISBN: 9780971764750

Either the physical book or a PDF that you are likely able to find online will work.

Course Schedule

The schedule below is subject to change. All due dates are listed in the Course Canvas page.

Module	Focus Of Class
 1	**What is GIS?:** • Course Introduction • History of GIS • Exploring online maps • Getting started with ArcGIS Online • Data Management
 2	**Elements of Mapping:** • Digital Data • Principles of Map Making • Cartography • Scale and Symbols and Marginalia • LOST (Legend, Orientation, Scale, Title)
 3	**Data Models and Map Reading:** • Data model types • Reading maps • Coordinate Systems • File Structures

 4	**Coordinate Reference Systems:** • Geodesy, Datums, Projections • Geographic Coordinate Systems (GCS) • Projected Coordinate Systems (PCS) • Coordinate conversions and uploading location data
 5	**Data Entry, Editing:** • GIS Database • Digitizing • Metadata • Demography
 6	**GNSS and Coordinate Surveys** • GPS and GNSS data collection • Receiver to map workflow **Midterm Work** • No new lab. Midterm work time
 7	**Managing Vector data:** • Importing data from spreadsheets and geocoding • Tools: Merging, appending, generalize, & dissolve • Discrete Data Formats (vector data formats) • Environment Settings **Midterm Gallery** • Map peer reviews and Gallery

8
	Managing Raster data:
	
	• Raster types (continuous, discrete, images)
	  ○ File formats (lossy and lossless)
	  ○ Bit depth & radiometric resolution
	  ○ Float vs integer / signed vs unsigned
	• Resolution, Pyramids, & Resampling
	• Raster coordinate systems and transformations
	• DEMs
	• Georeferencing
	• Share data into ArcGIS Group

 9	**Attribute data:** • Databases • Joins and cardinality • Table statistics • Field types and table editing • Queries and relates • Creating Domains

10
	Editing and Analysis:
	
	• Schema Design
	• Tracing
	• Creating Feature Classes
	  ○ Points, lines, polygons
	  ○ Sketching, Snapping, coincident boundaries
	• Vector data from imagery (buildings, logging, waterbodies, etc…)

 11	**Terrain Analysis:** • Slope and Aspect • Terrain analysis • Least Cost
 12	**Spatial Estimation, Interpolation:** • Regression • IDW • Nearest Neighbor

 13	**Spatial Models:** • Map Algebra (Raster Calculator) • Raster Boolean operators • Weighted overlays • Zonal Statistics
  14	**Final Project:** • Final project work • Future of GIS • Next steps

Weekly Expectations

Each Week will consist of:

- Lectures on module concepts
- Individual work time to complete Lab exercises. Lab exercise work through has been recorded and can be watched in class.
    - Students will be expected to submit individual labs but they can be worked on collectively. They will require out-of-class time to finish (Homework).
    - **Expect to spend at approximately 9 hours per week on coursework.**
- A canvas quiz on the weekly concepts

A final project will be completed utilizing concepts from the class. More information about the final project is available on the corresponding module page.

Assignments are due at 11:59 PM on the Thursday after they were assigned.

LATE policy: <u>A 5% deduction in points will be applied PER DAY the assignment is late up to a 50% reduction in grade.</u>

Credits & Grading

3 credits will be awarded upon successful completion of this course

You can not earn credits for both FORS 250 and GPHY 284

A : 93 – 100%, A- : 90 – 92.99%, B+ : 87 – 89.99%, B : 83 – 86.99%, B- : 80 – 82.99%, C+ : 77 – 79.99%, C : 73 – 76.99%, C- : 70 – 72.99%, D+ : 67 - 69.99%, D : 63 – 66.99%, D- : 60 - 62.99%, F : 59.99% and below

Course point breakdown (100 points. 1 point = 1% of final grade):

- 12 weekly labs worth 5 points each: 60 points possible
  - Some assignments will have 'Extra Credit' worth additional points.
  - Extra credit assignments will be in some of the answer keys you are provided after you submit your work. Not all answer keys will have extra credit options.
- 12 weekly online quizzes worth 1 point each: 12 points possible
  - Quizzes are focused on GIS concepts while the assignments are focused on applications.
  - You will have 2 attempts for each quiz.
- Midterm map project (10 points)
  - 7 points for producing a map with linked online content
  - 3 points for providing peer review of others projects
    - Note that you can't be assigned peer review unless you submit your own final project **on time**.
- A final project (18 points)
  - 1 point for posting your topic
  - 1 point for posting your data sources
  - 13 points for the project itself based on a provided rubric
  - 3 points for providing peer review of others projects
    - Note that you can't be assigned peer review unless you submit your own final project **on time**.

There are 100 points for the course. With the extra credit (contained within the answer keys), it is possible to get more than 100 points in this class. Late assignments will make obtaining credit much harder.

Assignments are due at 11:59 PM on the Thursday after they were assigned.

LATE policy: <u>A 5% deduction in points will be applied PER DAY the assignment is late up to a 50% reduction in grade.</u>

Homework

You can expect 4 hours of homework per week for this class (much more if you miss the in-person class). You can either work in the campus computer labs or install ArcGIS Pro on your own computer if it is powerful enough.

Computer specifications for ArcGIS Pro software.

ArcGIS downloading and installation instructions.

Campus Computer Labs:

Rooms 218 and 219 in Stone Hall are dedicated computer labs supporting coursework and research in geography, conservation, and climate science. Room 219 provides swipe access via Griz Card and connects internally to Room 218, allowing students to move between labs when permitted. Labs may be accessed outside scheduled class times, and Stone Hall is unlocked during regular business hours (8:00 AM–5:00 PM, Monday through Friday).
Students enrolled in this course will automatically receive card access. Students are expected to follow posted lab rules and maintain a respectful academic environment.

Other campus computer lab options are:

- Library main floor computers as well as some of the studio computers.
- Computer lab Native American 014

Responsibilities

We are entering a social contract together as we begin this academic semester. We each have responsibilities to one another. Responsibilities as a teacher and student. Some of these responsibilities are:

Teacher Responsibilities

- **Treat everyone in the room with respect.**
- **Maintain clear, unbiased, and unambiguous expectations for assessment.**
    - **Formative assignments** (e.g., weekly quizzes, in-class activities) are low-stakes opportunities for practice, correction, and discussion. They highlight important concepts and help you prepare for higher-stakes work.
    - **Summative assignments** (e.g., weekly labs, final project) carry the majority of points in the course. Strong performance on these is necessary to demonstrate subject mastery and receive credit.
- **Respond to student communication promptly.**
  Replies will be made within 48 hours (excluding weekends). Additional support will be provided during office hours.
- **Use Canvas announcements** to inform students of any schedule or assignment changes.
- **Foster an inclusive learning environment** that accommodates diverse learners.
  Please share any access needs with me, and feel free to move around or use silent fidgets during class.
- **Be committed to listening to student concerns** about the course, myself, or others, and address them respectfully and constructively.

Student Responsibilities

- **Treat everyone in the room with respect.**
  This includes engaging professionally, valuing diverse perspectives, and listening actively. Bigotry of any kind, or dismissive or demeaning comments about another person's identity (including race, ethnicity, gender, sexual orientation, cultural upbringing, disability, or other personal attributes) will not be tolerated.
- **Take responsibility for your own learning.**

  Metacognition (click the link for more info!) —"thinking about thinking"—means being aware of your learning needs and assessing your goals. Use campus resources (linked below) to support your success, and seek help when needed.

- **Stay current with course communications.**
  - Monitor Canvas announcements and messages, and ensure they are forwarded to your email if necessary.
  - Respond to direct communication within 48 hours (excluding weekends).
  - Familiarize yourself with the Canvas course structure.
- **Fully participate** in class activities and assignments.
- **Submit all work on time.**
- **Uphold academic honesty** as defined in the policy below.
- **Read all assignment feedback** carefully and use it to improve your work.
- **Provide constructive feedback** about the course, including what has worked well and what could be improved.

Note: Additional responsibilities may arise during the course. This list is not exhaustive. Please communicate any questions or concerns promptly.

Use of Artificial Intelligence (AI) Tools

AI tools such as ChatGPT, Copilot, or similar platforms are increasingly part of professional workflows in GIS, remote sensing, and cartography. In this course, you may use AI tools as a support for learning, provided that you follow these guidelines:

- **Permitted uses:** You may use AI tools to help brainstorm project ideas, debug R or Python code, clarify GIS concepts, or practice writing.
- **Prohibited uses:** You may not submit AI-generated maps, text, or code as your own without modification or acknowledgment. Copying and pasting AI outputs into assignments or projects without your own editing, verification, and synthesis is considered academic misconduct.

- **Attribution:** If you use AI in your work, include a brief note describing how it was used (e.g., “Used ChatGPT to troubleshoot ArcGIS Pro error messages and adapted the suggested Python snippet to my dataset”).
- **Responsibility:** You are responsible for verifying the accuracy of any AI output. These tools often produce incorrect or misleading results, especially with technical topics like coordinate systems, projections, and statistical methods.
- **Instructor discretion:** Certain labs or assignments may explicitly prohibit AI use. Any restrictions will be stated clearly in the assignment instructions.

The purpose of this policy is not to discourage AI use, but to ensure that it supports your learning and that your submitted work reflects your own understanding of GIS concepts and practices.

University Policies

Student Conduct Code and Community Standards

The Student Conduct Code at the University of Montana embodies and promotes honesty, integrity, accountability, rights, and responsibilities associated with constructive citizenship in our academic community. This Code describes expected standards of behavior for all students, including academic conduct and general conduct, and it outlines students' rights, responsibilities, and the campus processes for adjudicating alleged violations.

ODE Accommodations

The University of Montana assures equal access to instruction through collaboration between students with disabilities, instructors, and the Office for Disability Equity (ODE). If you anticipate or experience barriers based on disability, please contact the ODE at 406-243-2243 or ode@umontana.edu, or visit Office of Disability Equity website for more information.

Land Acknowledgement

The University of Montana acknowledges that we are in the aboriginal territories of the Salish and Kalispel people. Today, we honor the path they have always shown us in caring for this place for the generations to come.

Technology & Canvas Support

The University of Montana has designed a course for students to learn the basics of Canvas. UM Passport to Canvas is a self-paced online course to learn everything you need to know about using Canvas in your classes!

Canvas Resources for Students is another resource with curated content that includes videos, guides, and detailed instructions covering information on Assignments Discussions, Quizzes, and Grades.